

SEMESTER PROJECT PROPOSAL

Cryptocurrency Market Analytics using Apache Spark

Submitted By

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Project Phase: Phase 1

Planned Platform: Databricks Free Edition

Dashboard Tool: Power BI

GitHub Repository Link: <https://github.com/l242517/Crypto-Market-Analytics-24i-2516-24L-2517>

1. Project Introduction

For this project, we have selected cryptocurrency market analytics as our domain. The purpose of the project is to build a data pipeline that collects cryptocurrency market data, processes it using Apache Spark and finally displays useful results in a Power BI dashboard.

We will mainly work with Bitcoin, Ethereum and Solana. We are not trying to predict cryptocurrency prices or give investment advice. The purpose is to use real data for learning data ingestion, incremental loading, Spark transformations and the Bronze, Silver and Gold architecture.

2. Domain and Data Source

The data will be taken from the CoinGecko REST API. CoinGecko provides public cryptocurrency market data in JSON format. We downloaded the historical data using its market chart endpoint.

The endpoint used follows this format:

```
/api/v3/coins/{coin_id}/market_chart?vs_currency=usd&days=365
```

The coin ID is changed for each cryptocurrency. For example, bitcoin is used for Bitcoin, ethereum for Ethereum and solana for Solana. The response contains three main parts:

- prices - historical price values in US dollars
- market_caps - historical market capitalization values
- total_volumes - historical trading volume values

Each value is stored with a timestamp. This means the timestamp and the market value can later be converted into proper rows using Spark.

3. Full Load and Incremental Load

Full Load

The full load will be the first large load of historical data. For Phase 1, we downloaded 365 days of data separately for Bitcoin, Ethereum and Solana. These files will be treated as the starting dataset of the project.

The following full-load sample files have been prepared:

- bitcoin_full_load.json
- ethereum_full_load.json
- solana_full_load.json

Incremental Load

After the first full load, the pipeline should not download and process the complete history every day. It should only process data that is newer than the latest timestamp already stored.

For the Phase 1 sample, we downloaded recent seven-day data for the same three cryptocurrencies. These smaller files represent the type of payload that will arrive after the full load.

- bitcoin_incremental_load.json
- ethereum_incremental_load.json
- solana_incremental_load.json

During the actual implementation, we will compare the incoming timestamps with the maximum timestamp already present in the Silver table. Only newer records will be inserted. If the API returns an updated value for an existing timestamp, the matching record can be updated instead of creating a duplicate.

4. Data Sample, Volume and Frequency

All sample files are real JSON responses downloaded from the CoinGecko API. The full-load files contain the 365-day history, while the incremental sample files contain a smaller recent period.

The current dataset is small because only three cryptocurrencies have been selected. The measured full-load files are 37 KB, 38 KB and 37 KB, totaling about 112 KB. The incremental files are 17 KB, 18 KB and 18 KB, totaling about 53 KB, which is much smaller than the full files.

Our planned ingestion frequency is once per day. In Phase 2, we may increase the number of coins after confirming that the pipeline works correctly. We can expand from three coins to around 10 coins without changing the basic pipeline design. Starting with three coins will make testing easier and will reduce unnecessary use of free compute resources.

Estimated load plan:

Load type	Data included	Expected frequency
Full load	365-day history for the selected coins	One time at the start
Incremental load	Only new or recently updated observations	Daily

5. Security and PII

The downloaded data does not contain personal names, email addresses, phone numbers, IP addresses, passwords, wallet addresses or bank details. It only contains public market information such as timestamps, prices, market capitalization and trading volume. Therefore, our current dataset does not contain personally identifiable information.

Even though the data does not contain PII, we will still follow some basic security practices:

- Only required fields will be kept in the processed tables.
- An API key will not be uploaded to GitHub if one is used later.
- Unexpected fields will be checked before the data is moved to the Silver layer.
- We will not collect user accounts, wallet information or individual transaction records.

6. Planned Medallion Architecture

Bronze Layer

The Bronze layer will contain the raw JSON files exactly as received from the API. No major cleaning will be performed at this stage. We will also keep simple information such as the coin name, file name, load type and ingestion time so that we can identify where each file came from.

Silver Layer

The CoinGecko JSON stores price, market cap and volume in three separate arrays. In the Silver layer, Spark will flatten these arrays and combine their values using the coin ID and timestamp. The timestamps will be converted into readable dates and the numeric values will be cast to suitable data types.

The planned Silver table will contain these columns:

- coin_id
- coin_name
- observation_timestamp
- observation_date
- price_usd
- market_cap_usd
- total_volume_usd
- ingestion_timestamp

The main cleaning tasks will be removing duplicate records, checking null values, converting timestamps, checking numeric data types and keeping one record for each coin and timestamp.

Gold Layer

The Gold layer will contain tables that are easy to connect with Power BI. Instead of exposing raw JSON, these tables will contain calculated and summarized values.

- gold_daily_coin_metrics - daily price, market cap, volume and percentage change
- gold_coin_performance_summary - start price, latest price, minimum, maximum, average and overall change
- gold_market_comparison - comparison of the latest market cap and average volume of each coin
- gold_monthly_trends - monthly averages for price, market cap and volume

A simple dimensional model may also be used. It can contain a coin dimension, a date dimension and a fact table containing the daily market values.

7. Business Intelligence Dashboard

The final Power BI dashboard will allow the user to select a cryptocurrency and date range. The dashboard will mainly be used to compare the three selected coins and observe changes over time.

The dashboard will answer the following questions:

- How did the price of each cryptocurrency change over time?
- Which selected coin has the highest market capitalization?
- Which coin has the highest trading volume?
- Which coin had the highest percentage increase or decrease during the selected period?
- Did a major change in trading volume happen at the same time as a noticeable price change?

Planned dashboard visuals:

- KPI cards for latest price, latest market cap, latest volume and best-performing coin
- A line chart showing price movement over time
- A bar chart comparing market capitalization between coins
- A bar chart comparing trading volume
- A summary table showing starting price, ending price, minimum, maximum and percentage change

8. GitHub Repository and Version Control

A GitHub repository will be created for the project. Both group members will use the repository to store the proposal, sample data and future Spark notebooks. The repository link will be added to this proposal before submission.

Planned repository structure:

```
crypto-market-analytics/  
README.md  
data/full_load/  
    bitcoin_full_load.json  
    ethereum_full_load.json  
    solana_full_load.json  
data/incremental_load/  
    bitcoin_incremental_load.json  
    ethereum_incremental_load.json  
    solana_incremental_load.json  
docs/project_proposal.pdf  
notebooks/ (to be added in Phase 2)  
dashboard/ (to be added in Phase 3)
```

The README file will contain the project name, team member details, data source, project purpose and folder descriptions.

9. Tools and FinOps Plan

We plan to use Apache Spark with PySpark on Databricks Free Edition. Raw files will be stored in JSON format. The processed Silver and Gold data can be stored in Delta or Parquet format. Power

BI Desktop will be used for the final dashboard.

Since we are planning to work within free-tier limits, we will follow these steps:

- Start testing with only one coin and a small date range.
- Run the 365-day files only after the basic code works.
- Use incremental loading instead of downloading the whole history every day.
- Avoid creating unnecessary copies of the same data.
- Stop the compute resources when they are not being used.
- Increase the number of coins only if the free environment handles the three-coin pipeline properly.

This approach should keep the project manageable and prevent unnecessary use of storage and compute.

10. Expected Final Project

By the end of the project, we expect to have an end-to-end pipeline that first loads historical cryptocurrency data and then processes new data incrementally. The raw data will be kept in Bronze, cleaned data will be stored in Silver and business-level summaries will be created in Gold. The Power BI dashboard will use the Gold tables to display cryptocurrency market trends and comparisons.

The project will demonstrate REST API ingestion, full loading, incremental loading, Apache Spark transformations, the Medallion Architecture, simple dimensional modeling, Power BI reporting, GitHub version control and free-tier cost management.

11. References

1. CoinGecko API Documentation.
2. Apache Spark Documentation.
3. Databricks Documentation.
4. Microsoft Power BI Documentation.

Note before submission: Add the GitHub repository link and write the exact sizes of the downloaded JSON files after checking their file properties.